HEISENBERG-TWISTED MÖBIUS DYNAMICS: A NONCOMMUTATIVE DIRICHLET PRODUCT, AN OPERATOR ZETA INVERSE, AND EXACT POSITION–MOMENTUM EQUATIONS OF MOTION

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Remark 0.1 (Mathematical status and novelty status). This is Version 2 of an exploratory research notebook. The central change from Version 1 is structural: the canonical commutation relation is no longer placed next to an otherwise diagonal zeta multiplier. Instead it deforms the arithmetic product itself through the Weyl cocycle. All operator domains, convergence regions, and algebraic products used in the rigorous results are specified below.

The Weyl representation, the canonical commutation relation, Möbius inversion, and the standard arithmetic functions used as ingredients are standard. The *Heisenberg-twisted Dirichlet convolution*, the associated Möbius inverse coefficients $z_\vartheta(n)$, the prime-pair interference formula in

Theorem 6.1, and the specific Möbius Hamiltonian dynamics in Section 8 are introduced here as research-notebook constructions. No claim of literature priority is made in this version; a separate novelty audit would be required before any priority statement.

1. Introduction: what Version 2 changes

The first version produced a mathematically valid Fourier-multiplier model, but its Riemann and Heisenberg components were not dynamically coupled. In particular, a parameter σ was external to the canonical pair, a "Hamiltonian" of the form A^*A added no special dynamics, and plane waves of exact momentum existed whether or not an arithmetic symbol vanished. The resulting non-localization statement was therefore a standard consequence of $[Q, P] = i\hbar I$ appended to a separate zeta multiplier, rather than an arithmetic consequence of the Heisenberg algebra.

Version 2 removes that architecture. The starting point is the exact Schrödinger pair

$$[Q, P] = i\hbar I, \tag{1.1}$$

and its Weyl displacement operators. An integer n is assigned two arithmetically defined phase-space increments:

$$x_n = \ell \log n, \qquad y_n = \kappa \Omega(n), \tag{1.2}$$

where $\Omega(n)$ is the total number of prime factors of n, counted with multiplicity. The coordinate x_n measures multiplicative size; on the square-free support of μ, the integer $\Omega(n)$ is also the parity counter in

$$\mu(n) = (-1)^{\Omega(n)}. \tag{1.3}$$

Thus the two canonical directions encode two different arithmetic quantities already present in Möbius arithmetic.

The corresponding Weyl operators do not multiply according to the ordinary commutative law of Dirichlet convolution. The exact CCR produces a phase cocycle. This gives a new product $\star_\vartheta$ on arithmetic functions. Möbius is then inverted with respect to this deformed product, producing coefficients $z_\vartheta(n)$ which equal 1 in the commuting limit but acquire explicit cross-prime interference for nonzero Heisenberg area.

The central rigorous formula proved below is

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1 = 1 - 4\sin^2\left(\frac{\vartheta}{4}\log\frac{q}{p}\right) \tag{1.4}$$

for distinct primes p, q, whereas

$$z_\vartheta(p^r) = 1 \tag{1.5}$$

for every prime power. Equation (1.4) is not a change of notation for the classical identity $\mu * 1 = \delta_1$: it is the coefficient forced by inverting Möbius in the noncommutative arithmetic algebra generated by the Weyl form of Heisenberg's relation.

A genuine time evolution is then defined from the self-adjoint Möbius Hamiltonian $H_{\mu,\sigma,\vartheta}$. Position and momentum are explicitly Q and P, and their exact Heisenberg equations are derived in Theorem 8.6. No variance or measurement-error interpretation is required anywhere in that theorem.

Warning 1.1 (What is not claimed). Version 2 does not claim a proof or a new equivalence of the Riemann Hypothesis. The ordinary zeta function is recovered at the commuting boundary $\vartheta = 0$. For $\vartheta \neq 0$ the inverse is a genuinely deformed zeta object and its continuation and zero set are open problems. The purpose of this version is to produce a non-tautological coupling between Heisenberg noncommutativity and Möbius inversion before asking what, if anything, this coupling implies about the classical zeros.

2. Exact canonical kinematics

2.1. Hilbert space, position, and momentum.

Definition 2.1 (Hilbert space)**.** Fix $\hbar > 0$ and set

$$\mathcal{H} = L^2(\mathbb{R}, dq) \tag{2.1}$$

with its standard complex Hilbert-space inner product.

Definition 2.2 (Position)**.** The position operator Q is

$$(Q\psi)(q) = q\psi(q) \tag{2.2}$$

with maximal domain

$$\mathcal{D}(Q) = \{\psi \in L^2(\mathbb{R}) : q\psi(q) \in L^2(\mathbb{R})\}. \tag{2.3}$$

Definition 2.3 (Momentum)**.** The momentum operator P is

$$(P\psi)(q) = -i\hbar\,\psi'(q) \tag{2.4}$$

with domain

$$\mathcal{D}(P) = H^1(\mathbb{R}). \tag{2.5}$$

Proposition 2.4 (Self-adjointness and common core)**.** *The operators Q and P are self-adjoint. The Schwartz space $\mathcal{S}(\mathbb{R})$ is a common invariant core for both.*

Proof. Q is the maximal multiplication operator by the real function q. Under the unitary Fourier transform with convention

$$(\mathcal{F}\psi)(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{-ipq/\hbar}\psi(q)\,dq, \tag{2.6}$$

P is unitarily equivalent to multiplication by the real variable p. The core statement follows by standard cutoff and mollification arguments. See, for example, [3]. □

Proposition 2.5 (Exact canonical commutation relation)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$[Q, P]\psi = i\hbar\psi. \tag{2.7}$$

Proof. Direct differentiation gives

$$(QP\psi)(q) = -i\hbar q\psi'(q), \tag{2.8}$$

$$(PQ\psi)(q) = -i\hbar\psi(q) - i\hbar q\psi'(q). \tag{2.9}$$

Subtracting (2.9) from (2.8) proves (2.7). □

2.2. Exact Heisenberg impossibility, without variances.

Theorem 2.6 (No simultaneous exact position and momentum)**.** *There is no nonzero state with simultaneous exact position and exact momentum. More precisely, both of the following are impossible for real q_0, p_0:*

(1) a nonzero vector ψ in a common invariant domain satisfying

$$Q\psi = q_0\psi, \qquad P\psi = p_0\psi; \tag{2.10}$$

(2) a nonzero tempered distribution $T \in \mathcal{S}'(\mathbb{R})$ satisfying, in the distributional sense,

$$qT = q_0T, \qquad -i\hbar T' = p_0T. \tag{2.11}$$

Proof. For vectors, (2.10) implies

$$[Q,P]\psi = 0, \tag{2.12}$$

whereas (2.7) gives

$$[Q,P]\psi = i\hbar\psi. \tag{2.13}$$

Thus $\psi = 0$.

For distributions define directly

$$QT = qT, \qquad PT = -i\hbar T'. \tag{2.14}$$

The distributional Leibniz rule gives

$$[Q,P]T = i\hbar T. \tag{2.15}$$

If (2.11) held, the left side would also vanish, so $T = 0$. □

Remark 2.7. Theorem 2.6 is the exact algebraic content used in this notebook when the word "Heisenberg" is invoked. It is not a statement about experimental error, measurement disturbance, or a product of standard deviations.

3. Weyl displacements and the arithmetic phase space

3.1. **Weyl operators.**

Definition 3.1 (Weyl displacement)**.** For $(x,y) \in \mathbb{R}^2$, define

$$W(x,y) = \exp\left(\frac{i}{\hbar}(yQ - xP)\right). \tag{3.1}$$

This is a unitary operator on all of $\mathcal{H}$.

Proposition 3.2 (Coordinate action)**.** *For $\psi \in L^2(\mathbb{R})$,*

$$(W(x,y)\psi)(q) = \exp\left(\frac{i}{\hbar}y\left(q - \frac{x}{2}\right)\right)\psi(q-x) \tag{3.2}$$

almost everywhere.

Proof. Since $[iyQ/\hbar, -ixP/\hbar]$ is scalar, the Baker–Campbell–Hausdorff formula gives

$$W(x,y) = e^{-ixy/(2\hbar)}e^{iyQ/\hbar}e^{-ixP/\hbar}. \tag{3.3}$$

The last factor translates by x and the middle factor multiplies by $e^{iyq/\hbar}$, yielding (3.2). □

Proposition 3.3 (Weyl form of the CCR)**.** *Let $z = (x,y)$ and $z' = (x',y')$. Define*

$$\omega(z,z') = yx' - xy'. \tag{3.4}$$

Then

$$W(z)W(z') = \exp\left(\frac{i}{2\hbar}\omega(z,z')\right)W(z+z'). \tag{3.5}$$

Consequently,

$$[W(z),W(z')] = 2i\sin\left(\frac{\omega(z,z')}{2\hbar}\right)W(z+z'). \tag{3.6}$$

Proof. For

$$A_z = \frac{i}{\hbar}(yQ - xP), \tag{3.7}$$

the CCR gives

$$[A_z, A_{z'}] = \frac{i}{\hbar}\omega(z, z')I. \tag{3.8}$$

This commutator is central, so the Baker–Campbell–Hausdorff series stops after the commutator term and gives (3.5). Reversing z and z' changes the sign of ω, giving (3.6). The Weyl relations are standard; see [2]. □

Proposition 3.4 (Position and momentum displacement equations)**.** *On* $\mathcal{S}(\mathbb{R})$*,*

$$[Q, W(x, y)] = xW(x, y), \tag{3.9}$$

$$[P, W(x, y)] = yW(x, y). \tag{3.10}$$

Proof. Both identities follow by differentiating (3.2), or directly from (2.7) because the relevant commutators with the exponent are scalar. □

3.2. **Arithmetic embedding.**

Definition 3.5 (The additive prime-factor counter)**.** For

$$n = \prod_p p^{v_p(n)}, \tag{3.11}$$

define

$$\Omega(n) = \sum_p v_p(n). \tag{3.12}$$

Then

$$\Omega(mn) = \Omega(m) + \Omega(n). \tag{3.13}$$

Definition 3.6 (Arithmetic phase-space map)**.** Fix two positive scales $\ell > 0$ and $\kappa > 0$. For $n \in \mathbb{N}$, put

$$x_n = \ell \log n, \tag{3.14}$$

$$y_n = \kappa\Omega(n), \tag{3.15}$$

$$z_n = (x_n, y_n). \tag{3.16}$$

Define the dimensionless Heisenberg area parameter

$$\vartheta = \frac{\ell\kappa}{\hbar}. \tag{3.17}$$

Remark 3.7 (Why these two coordinates). The first coordinate is forced by multiplicative size because $\log(mn) = \log m + \log n$. The second coordinate is chosen as the simplest integer-valued completely additive function that directly controls the Möbius sign on square-free integers:

$$\mu(n) \neq 0 \quad \Longrightarrow \quad \mu(n) = (-1)^{\Omega(n)}. \tag{3.18}$$

The construction can be generalized by replacing Ω with another completely additive arithmetic function, but the present notebook fixes Ω throughout.

Definition 3.8 (Arithmetic Weyl operators)**.** For $n \in \mathbb{N}$, define

$$W_n = W(x_n, y_n). \tag{3.19}$$

Theorem 3.9 (Exact Heisenberg cocycle on the integers)**.** *For $m, n \in \mathbb{N}$,*

$$W_m W_n = \chi_\vartheta(m,n) W_{mn}, \tag{3.20}$$

where

$$\chi_\vartheta(m,n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right]. \tag{3.21}$$

Moreover,

$$[W_m, W_n] = 2i\sin\left[\frac{\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right] W_{mn}. \tag{3.22}$$

Proof. Complete additivity gives

$$z_{mn} = z_m + z_n. \tag{3.23}$$

Substitution of (3.14)–(3.17) into (3.5) gives (3.20); (3.22) follows from (3.6). □

Corollary 3.10 (Prime commutator)**.** *For distinct primes p, q,*

$$[W_p, W_q] = 2i\sin\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) W_{pq}. \tag{3.24}$$

Thus the prime operators commute only at the resonant values

$$\vartheta \log\frac{q}{p} \in 2\pi\mathbb{Z}. \tag{3.25}$$

Remark 3.11 (Where Heisenberg enters nontrivially). Equation (3.20) is the first point where Version 2 ceases to be ordinary Dirichlet algebra in operator notation. If the canonical pair were commuting, the Weyl cocycle would be 1 and the arithmetic product below would reduce to ordinary Dirichlet convolution. For nonzero ϑ, the CCR creates order-dependent phases between different arithmetic factors.

4. The Heisenberg-twisted Dirichlet algebra

4.1. **Definition and associativity.**

Definition 4.1 (Heisenberg-twisted Dirichlet convolution)**.** For arithmetic functions $f, g : \mathbb{N} \to \mathbb{C}$, define

$$(f \star_\vartheta g)(k) = \sum_{d|k} f(d)\,g(k/d)\,\chi_\vartheta(d, k/d). \tag{4.1}$$

The sum is finite for every k.

Lemma 4.2 (Cocycle identity)**.** *For all $a, b, c \in \mathbb{N}$,*

$$\chi_\vartheta(a,b)\chi_\vartheta(ab,c) = \chi_\vartheta(b,c)\chi_\vartheta(a,bc). \tag{4.2}$$

Proof. The exponent in (3.21) is the alternating bilinear form obtained from the additive map $n \mapsto z_n$. Expanding both sides using $\log(ab) = \log a + \log b$ and $\Omega(ab) = \Omega(a) + \Omega(b)$ gives the same exponent. □

Proposition 4.3 (Associative noncommutative arithmetic algebra)**.** *The product $\star_\vartheta$ is associative and has identity δ_1, where*

$$\delta_1(1) = 1, \qquad \delta_1(n) = 0 \quad (n > 1). \tag{4.3}$$

For $\vartheta \neq 0$ it is generally noncommutative.

Proof. Associativity follows from Lemma 4.2 after grouping the finite divisor sums by triples $abc = k$. The identity property follows from $\chi_\vartheta(1,n) = \chi_\vartheta(n,1) = 1$. Noncommutativity follows from Corollary 3.10; equivalently,

$$(\delta_p \star_\vartheta \delta_q)(pq) = \chi_\vartheta(p,q) \tag{4.4}$$

whereas

$$(\delta_q \star_\vartheta \delta_p)(pq) = \chi_\vartheta(q,p) = \overline{\chi_\vartheta(p,q)}. \tag{4.5}$$

□

4.2. **Weighted Banach algebra and operator representation.**

Definition 4.4 (Weighted arithmetic Banach space)**.** For $\sigma > 0$, set

$$\mathcal{A}_\sigma = \left\{ f : \mathbb{N} \to \mathbb{C} : \|f\|_\sigma := \sum_{n\geq 1} \frac{|f(n)|}{n^\sigma} < \infty \right\}. \tag{4.6}$$

Proposition 4.5 (Banach-algebra estimate)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|f \star_\vartheta g\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma. \tag{4.7}$$

Hence $(\mathcal{A}_\sigma, \star_\vartheta)$ is a unital Banach algebra.

Proof. Since $|\chi_\vartheta| = 1$,

$$\|f \star_\vartheta g\|_\sigma \leq \sum_{k\geq 1} \frac{1}{k^\sigma} \sum_{d|k} |f(d)||g(k/d)| \tag{4.8}$$

$$= \sum_{d,e\geq 1} \frac{|f(d)|}{d^\sigma} \frac{|g(e)|}{e^\sigma} \tag{4.9}$$

$$= \|f\|_\sigma \|g\|_\sigma. \tag{4.10}$$

Completeness is the completeness of weighted ℓ^1. □

Definition 4.6 (Weyl–Dirichlet representation)**.** For $f \in \mathcal{A}_\sigma$, define

$$\Pi_{\sigma,\vartheta}(f) = \sum_{n\geq 1} \frac{f(n)}{n^\sigma} W_n, \tag{4.11}$$

where the series converges absolutely in operator norm.

Proposition 4.7 (Representation theorem)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|\Pi_{\sigma,\vartheta}(f)\| \leq \|f\|_\sigma, \tag{4.12}$$

$$\Pi_{\sigma,\vartheta}(f)\Pi_{\sigma,\vartheta}(g) = \Pi_{\sigma,\vartheta}(f \star_\vartheta g). \tag{4.13}$$

Proof. The norm bound follows from $\|W_n\| = 1$. Absolute norm convergence permits multiplication and regrouping of the double series. The coefficient of W_k is exactly (4.1) by Theorem 3.9. □

5. Möbius inversion deformed by the CCR

5.1. **The Möbius operator.**

Definition 5.1 (Möbius operator)**.** For $\sigma > 1$, define

$$\mathcal{M}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(\mu) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} W_n. \tag{5.1}$$

The series converges absolutely in operator norm, so $\mathcal{M}_{\sigma,\vartheta} \in \mathcal{B}(\mathcal{H})$ and

$$\mathcal{D}(\mathcal{M}_{\sigma,\vartheta}) = \mathcal{H}. \tag{5.2}$$

Remark 5.2 (Concrete coordinate action). For $\psi \in L^2(\mathbb{R})$,

$$(\mathcal{M}_{\sigma,\vartheta}\psi)(q) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} \exp\left[\frac{i}{\hbar} y_n \left(q - \frac{x_n}{2}\right)\right] \psi(q - x_n), \tag{5.3}$$

with convergence in L^2. Thus the operator is a Möbius-weighted superposition of simultaneous translations in position and momentum; it is not a Fourier multiplier when $\kappa \neq 0$.

5.2. **Formal twisted inverse.**

Proposition 5.3 (Existence and uniqueness of arithmetic inverses)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ satisfy $f(1) \neq 0$. Then there exists a unique arithmetic function g such that*

$$f \star_\vartheta g = g \star_\vartheta f = \delta_1. \tag{5.4}$$

Its coefficients are determined recursively from $g(1) = f(1)^{-1}$.

Proof. For $n > 1$, the equation $(f \star_\vartheta g)(n) = 0$ contains the term $f(1)g(n)$ and otherwise only values $g(d)$ with $d < n$. This determines a unique left inverse recursively. The same recursion gives a unique right inverse. In an associative unital algebra, a left and right inverse coincide. □

Definition 5.4 (Heisenberg-zeta coefficients)**.** Let $z_\vartheta : \mathbb{N} \to \mathbb{C}$ be the unique two-sided inverse of μ for $\star_\vartheta$:

$$\mu \star_\vartheta z_\vartheta = z_\vartheta \star_\vartheta \mu = \delta_1. \tag{5.5}$$

Thus

$$z_\vartheta(1) = 1, \tag{5.6}$$

and for $n > 1$,

$$z_\vartheta(n) = -\sum_{\substack{d|n \\ d>1}} \mu(d) z_\vartheta(n/d) \chi_\vartheta(d, n/d). \tag{5.7}$$

Proposition 5.5 (Commuting boundary)**.** *At $\vartheta = 0$,*

$$z_0(n) = 1 \tag{5.8}$$

for every $n \geq 1$.

Proof. When $\vartheta = 0$, $\chi_0 \equiv 1$ and $\star_\vartheta$ is ordinary Dirichlet convolution. Equation (5.5) becomes $\mu * z_0 = \delta_1$. Classical Möbius inversion gives $z_0 = \mathbf{1}$. See [4]. □

5.3. **A bounded inverse in a rigorous half-plane.**

Theorem 5.6 (Uniform Neumann region)**.** *For every real ϑ,*

$$z_\vartheta \in \mathcal{A}_2. \tag{5.9}$$

Consequently, for every $\sigma \geq 2$ the operator series

$$\mathcal{Z}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(z_\vartheta) = \sum_{n\geq 1} \frac{z_\vartheta(n)}{n^\sigma} W_n \tag{5.10}$$

converges absolutely in operator norm and satisfies

$$\mathcal{M}_{\sigma,\vartheta}\mathcal{Z}_{\sigma,\vartheta} = \mathcal{Z}_{\sigma,\vartheta}\mathcal{M}_{\sigma,\vartheta} = I. \tag{5.11}$$

Proof. Write

$$\mu = \delta_1 + h, \qquad h(1) = 0, \qquad h(n) = \mu(n)\ (n > 1). \tag{5.12}$$

Since $|\mu(n)| = \mu(n)^2$,

$$\|h\|_2 = \sum_{n\geq 2} \frac{\mu(n)^2}{n^2} \tag{5.13}$$

$$= \frac{\zeta(2)}{\zeta(4)} - 1 \tag{5.14}$$

$$= \frac{15}{\pi^2} - 1 < 1. \tag{5.15}$$

The equality in (5.14) is the standard Dirichlet series for square-free integers. Proposition 4.5 now permits the Neumann series

$$(\delta_1 + h)^{-1} = \sum_{r=0}^{\infty} (-h)^{\star_\vartheta r} \tag{5.16}$$

in $\mathcal{A}_2$. By uniqueness of the arithmetic inverse, this inverse is z_ϑ. Since $\mathcal{A}_2 \subseteq \mathcal{A}_\sigma$ for $\sigma \geq 2$, Proposition 4.7 gives (5.11). □

Warning 5.7 (Exact border of the proved inverse). The Möbius operator (5.1) exists as a bounded operator for every $\sigma > 1$. The argument above proves invertibility uniformly in ϑ only for $\sigma \geq 2$. No claim of invertibility for $1 < \sigma < 2$ is made here.

6. The first Heisenberg arithmetic law

Theorem 6.1 (Prime rays are rigid; distinct primes interfere)**.** *Let p, q be distinct primes and let $r \geq 1$. Then*

$$z_\vartheta(p^r) = 1, \tag{6.1}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1, \tag{6.2}$$

$$z_\vartheta(pq) - 1 = -4\sin^2\left(\frac{\vartheta}{4}\log\frac{q}{p}\right). \tag{6.3}$$

Proof. For p^r, only the divisors 1 and p contribute to the recursion because $\mu(p^j) = 0$ for $j \geq 2$. Moreover,

$$\Omega(p)\log(p^{r-1}) - \log p\,\Omega(p^{r-1}) = 0, \tag{6.4}$$

so $\chi_\vartheta(p, p^{r-1}) = 1$. Equation (5.7) gives

$$z_\vartheta(p^r) = z_\vartheta(p^{r-1}), \tag{6.5}$$

and (6.1) follows from $z_\vartheta(1) = 1$.

For pq, the four divisors $1, p, q, pq$ give

$$0 = z_\vartheta(pq) - \chi_\vartheta(p, q) - \chi_\vartheta(q, p) + 1. \tag{6.6}$$

Since

$$\chi_\vartheta(p, q) = \exp\left(\frac{i\vartheta}{2}\log\frac{q}{p}\right), \tag{6.7}$$

and $\chi_\vartheta(q, p)$ is its complex conjugate, (6.2) follows. The identity (6.3) is the elementary formula $2(\cos u - 1) = -4\sin^2(u/2)$. □

Corollary 6.2 (Loss of ordinary multiplicativity)**.** *Generically,*

$$z_\vartheta(pq) \neq z_\vartheta(p)z_\vartheta(q) = 1. \tag{6.8}$$

Thus the Heisenberg deformation preserves every one-prime ray but destroys ordinary multiplicativity through cross-prime interference.

Corollary 6.3 (Small-area expansion)**.** *For fixed distinct primes p, q,*

$$z_\vartheta(pq) = 1 - \frac{\vartheta^2}{4}\left(\log\frac{q}{p}\right)^2 + O(\vartheta^4) \qquad (\vartheta \to 0). \tag{6.9}$$

There is no linear correction.

Proposition 6.4 (Evenness of the full inverse)**.** *For every $n \geq 1$,*

$$z_{-\vartheta}(n) = z_\vartheta(n). \tag{6.10}$$

Proof. Changing ϑ to $-\vartheta$ reverses the cocycle:

$$(f \star_{-\vartheta} g)(n) = (g \star_\vartheta f)(n). \tag{6.11}$$

Since z_ϑ is a two-sided inverse of μ for $\star_\vartheta$, it is a two-sided inverse of μ for the opposite product $\star_{-\vartheta}$. Uniqueness of the inverse gives (6.10). □

Remark 6.5 (Why this is the non-tautological step). At $\vartheta = 0$ the inverse coefficients are identically 1. At nonzero ϑ, equation (6.3) creates a new arithmetic coefficient depending on the logarithmic separation of two primes. This dependence is forced by the symplectic area in the Weyl product and disappears if the CCR is removed. It is therefore the first result in the notebook where Heisenberg noncommutativity changes the arithmetic inverse itself rather than merely constraining an unrelated state space.

7. The Heisenberg-deformed zeta series

Definition 7.1 (Heisenberg-deformed zeta series)**.** For $\Re s \geq 2$, define

$$\zeta_{\mathrm{H}}(s;\vartheta) = \sum_{n\geq 1} \frac{z_\vartheta(n)}{n^s}. \tag{7.1}$$

Theorem 5.6 guarantees absolute convergence on the closed half-plane $\Re s \geq 2$.

Proposition 7.2 (Classical boundary)**.** *For $\Re s > 1$,*

$$\zeta_{\mathrm{H}}(s;0) = \zeta(s). \tag{7.2}$$

Proof. By Proposition 5.5, $z_0(n) = 1$ for all n. Substitution into (7.1) gives the ordinary Dirichlet series for $\zeta(s)$. □

Proposition 7.3 (Even Heisenberg deformation)**.** *For $\Re s \geq 2$,*

$$\zeta_{\mathrm{H}}(s;-\vartheta) = \zeta_{\mathrm{H}}(s;\vartheta). \tag{7.3}$$

Proof. This follows termwise from Proposition 6.4 and absolute convergence. □

Warning 7.4 (No analytic continuation is asserted). The definition (7.1) is presently rigorous only in the half-plane furnished by Theorem 5.6. No meromorphic continuation, functional equation, Euler product, or critical-strip theory for $\zeta_{\mathrm{H}}(s;\vartheta)$ is asserted here.

Proposition 7.5 (Conditional first-order rigidity of simple zeros)**.** *Assume that, for some simple zero ρ of the ordinary zeta function, there exists a function $F(s,\vartheta)$ analytic in a neighbourhood of $(\rho,0)$ such that*

$$F(s,0)=\zeta(s), \tag{7.4}$$

$$F(s,-\vartheta)=F(s,\vartheta), \tag{7.5}$$

and that F agrees with ζ_{H} wherever both are defined. Then the local zero branch $\rho(\vartheta)$ through ρ satisfies

$$\rho'(0)=0. \tag{7.6}$$

Proof. The implicit-function theorem applies because $\zeta'(\rho)\neq 0$. Evenness (7.5) makes the unique local zero branch even, so its first derivative at the origin vanishes. □

Warning 7.6. Proposition 7.5 is conditional on an analytic continuation in both variables and does not supply such a continuation. It is included because it identifies a precise way in which the exact $\vartheta\mapsto-\vartheta$ symmetry would constrain the motion of classical Riemann zeros under the Heisenberg deformation if that continuation exists.

8. A genuine Möbius Hamiltonian

8.1. **Definition and domain.**

Definition 8.1 (Cosine and sine Weyl quadratures)**.** For $n\geq 1$, set

$$C_n=\frac{W_n+W_n^*}{2}, \tag{8.1}$$

$$J_n=\frac{W_n-W_n^*}{2i}. \tag{8.2}$$

Both are bounded self-adjoint operators with norm at most 1.

Definition 8.2 (Möbius Hamiltonian)**.** Fix an energy scale $E_0>0$ and $\sigma>1$. Define

$$H_{\mu,\sigma,\vartheta}=E_0\,(\operatorname{Re}\mathcal{M}_{\sigma,\vartheta}-I)=E_0\sum_{n\geq 2}\frac{\mu(n)}{n^{\sigma}}C_n. \tag{8.3}$$

The series converges absolutely in operator norm.

Proposition 8.3 (Self-adjoint bounded dynamics)**.** *For $\sigma>1$, $H_{\mu,\sigma,\vartheta}$ is bounded and self-adjoint on all of $\mathcal{H}$:*

$$\mathcal{D}(H_{\mu,\sigma,\vartheta})=\mathcal{H}. \tag{8.4}$$

It therefore generates the unitary group

$$U(t)=\exp\!\left(-\frac{i}{\hbar}tH_{\mu,\sigma,\vartheta}\right). \tag{8.5}$$

Proof. The norm series in (8.3) is dominated by $E_0\sum_{n\geq 2}|\mu(n)|n^{-\sigma}$, which converges for $\sigma>1$. Every C_n is self-adjoint, hence so is the norm limit. The spectral theorem then defines (8.5). □

Proposition 8.4 (Explicit Schrödinger equation)**.** *For $\psi_0\in\mathcal{H}$, the evolution $\psi(t)=U(t)\psi_0$ satisfies*

$$i\hbar\,\partial_t\psi(t,q)=(H_{\mu,\sigma,\vartheta}\psi(t))(q) \tag{8.6}$$

in the strong L^2 sense, where

$$(H_{\mu,\sigma,\vartheta}\psi)(q) = \frac{E_0}{2}\sum_{n\geq 2}\frac{\mu(n)}{n^\sigma}\Bigg[e^{\frac{i}{\hbar}y_n(q-x_n/2)}\psi(q-x_n) + e^{-\frac{i}{\hbar}y_n(q+x_n/2)}\psi(q+x_n)\Bigg]. \tag{8.7}$$

Proof. Equation (8.7) follows from Proposition 3.2 applied to W_n and W_n^*. The series converges in operator norm, so the bounded Hamiltonian gives a strong Schrödinger evolution for every initial vector. □

8.2. **Exact equations of motion for position and momentum.**

Lemma 8.5 (Bounded first commutators)**.** *For $\sigma > 1$, the series*

$$\sum_{n\geq 2}\frac{|\mu(n)|x_n}{n^\sigma}, \tag{8.8}$$

$$\sum_{n\geq 2}\frac{|\mu(n)|y_n}{n^\sigma} \tag{8.9}$$

converge. Hence $[H_{\mu,\sigma,\vartheta}, Q]$ and $[H_{\mu,\sigma,\vartheta}, P]$, initially defined on $\mathcal{S}(\mathbb{R})$, extend to bounded operators.

Proof. The first series is bounded by a constant multiple of $\sum n^{-\sigma}\log n$. Since

$$\Omega(n) \leq \frac{\log n}{\log 2}, \tag{8.10}$$

the second is bounded by another constant multiple of the same convergent series. □

Theorem 8.6 (Exact Heisenberg equations of motion)**.** *Let*

$$Q(t) = U(t)^* Q U(t), \tag{8.11}$$

$$P(t) = U(t)^* P U(t). \tag{8.12}$$

On the common invariant core, and equivalently in the associated quadratic forms, the Heisenberg equations are

$$\dot{Q}(t) = \frac{E_0}{\hbar}U(t)^*\left[\sum_{n\geq 2}\frac{\mu(n)x_n}{n^\sigma}J_n\right]U(t), \tag{8.13}$$

$$\dot{P}(t) = \frac{E_0}{\hbar}U(t)^*\left[\sum_{n\geq 2}\frac{\mu(n)y_n}{n^\sigma}J_n\right]U(t). \tag{8.14}$$

At $t = 0$ this becomes

$$\dot{Q}(0) = \frac{E_0\ell}{\hbar}\sum_{n\geq 2}\frac{\mu(n)\log n}{n^\sigma}J_n, \tag{8.15}$$

$$\dot{P}(0) = \frac{E_0\kappa}{\hbar}\sum_{n\geq 2}\frac{\mu(n)\Omega(n)}{n^\sigma}J_n. \tag{8.16}$$

Proof. The time-independent Heisenberg equation is

$$\dot{A}(t) = \frac{i}{\hbar}U(t)^*[H_{\mu,\sigma,\vartheta}, A]U(t). \tag{8.17}$$

From Proposition 3.4,

$$[Q, C_n] = ix_n J_n, \tag{8.18}$$

$$[P, C_n] = iy_n J_n. \tag{8.19}$$

Therefore

$$[H_{\mu,\sigma,\vartheta}, Q] = -iE_0 \sum_{n\geq 2} \frac{\mu(n)x_n}{n^\sigma} J_n, \tag{8.20}$$

$$[H_{\mu,\sigma,\vartheta}, P] = -iE_0 \sum_{n\geq 2} \frac{\mu(n)y_n}{n^\sigma} J_n. \tag{8.21}$$

Lemma 8.5 justifies the sums as bounded operators. Substitution into (8.17) gives (8.13) and (8.14). □

Remark 8.7 (Physical and arithmetic meaning of the two equations). The position variable is Q, the logarithmic-scale coordinate. The momentum variable is $P = -i\hbar\partial_q$, the generator conjugate to that coordinate. Multiplication by an integer n is represented by a phase-space displacement of size $\ell \log n$ in Q and $\kappa\Omega(n)$ in P. Equations (8.15) and (8.16) therefore say that the Möbius Hamiltonian drives position by logarithmic size and momentum by prime-factor count, with the same quantum current operators J_n.

8.3. **The CCR is visible in the dynamics.**

Proposition 8.8 (Pairwise dynamical noncommutativity)**.** *Extend* log *and* Ω *additively to positive rationals and define* W_r *by* (3.19) *for* $r \in \mathbb{Q}_{>0}$*. For* $m, n \in \mathbb{N}$*, put*

$$\Phi_{m,n} = \vartheta\big(\Omega(m)\log n - \log m\,\Omega(n)\big). \tag{8.22}$$

Then

$$[C_m, J_n] = \frac{1}{2}\sin\!\left(\frac{\Phi_{m,n}}{2}\right)\Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{8.23}$$

Proof. Expand C_m and J_n using (8.1)–(8.2). Apply (3.6) to the four pairs (z_m, z_n), $(z_m, -z_n)$, $(-z_m, z_n)$, and $(-z_m, -z_n)$. □

Corollary 8.9 (Second-order arithmetic force)**.** *At* $t = 0$*,*

$$\ddot{Q}(0) = \frac{iE_0^2}{2\hbar^2} \sum_{m,n\geq 2} \frac{\mu(m)\mu(n)x_n}{(mn)^\sigma} \sin\!\left(\frac{\Phi_{m,n}}{2}\right) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big), \tag{8.24}$$

and the analogous momentum equation is

$$\ddot{P}(0) = \frac{iE_0^2}{2\hbar^2} \sum_{m,n\geq 2} \frac{\mu(m)\mu(n)y_n}{(mn)^\sigma} \sin\!\left(\frac{\Phi_{m,n}}{2}\right) \times \Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{8.25}$$

The double sums converge in operator norm for $\sigma > 1$*.*

Proof. Differentiate (8.13) and (8.14) once more and use Proposition 8.8. Polynomial logarithmic weights are summable against $(mn)^{-\sigma}$ for $\sigma > 1$. □

Corollary 8.10 (The interaction disappears with the Weyl cocycle)**.** *If* $\vartheta = 0$*, then*

$$\Phi_{m,n} = 0 \tag{8.26}$$

for all m, n*, and all pairwise terms in* (8.24)–(8.25) *vanish. Thus the second-order cross-arithmetic force displayed above is generated entirely by the exact Heisenberg cocycle.*

9. Exact non-localization inside the arithmetic dynamics

Proposition 9.1 (Schwartz invariance and distributional extension)**.** *For $\sigma > 1$, the Hamiltonian $H_{\mu,\sigma,\vartheta}$ maps $\mathcal{S}(\mathbb{R})$ continuously into itself. Consequently its coordinate formula* (8.7) *extends termwise to tempered distributions. For every fixed $T \in \mathcal{S}'(\mathbb{R})$ the resulting series converges in $\mathcal{S}'(\mathbb{R})$.*

Proof. For every Schwartz seminorm $p_{a,b}(\phi) = \sup_q |q^a \phi^{(b)}(q)|$, the coordinate action (3.2) gives a bound of the form

$$p_{a,b}(W_n\phi) \le C_{a,b}(1 + |x_n| + |y_n|)^{N_{a,b}} \sum_{j\le b} p_{a+j,b-j}(\phi), \tag{9.1}$$

for some finite integer $N_{a,b}$. Since x_n and y_n grow at most logarithmically and

$$\sum_{n\ge 2} n^{-\sigma}(1 + \log n)^N < \infty \qquad (\sigma > 1) \tag{9.2}$$

for every fixed N, the Hamiltonian series converges in every Schwartz seminorm. Translation and multiplication by a smooth phase act continuously on $\mathcal{S}'$; the same logarithmic summability yields convergence after applying the series to any fixed tempered distribution. □

Proposition 9.2 (No exact position point is a stationary orbital)**.** *Let $\sigma > 1$, $\ell > 0$, and $E_0 > 0$. For no $q_0 \in \mathbb{R}$ and no scalar $E \in \mathbb{C}$ does the tempered distribution δ_{q_0} satisfy*

$$H_{\mu,\sigma,\vartheta}\delta_{q_0} = E\delta_{q_0}. \tag{9.3}$$

Proof. By (3.2), $W_n\delta_{q_0}$ is a nonzero scalar multiple of $\delta_{q_0+x_n}$ and $W_n^*\delta_{q_0}$ is a nonzero scalar multiple of $\delta_{q_0-x_n}$. In particular, the $n = 2$ term of (8.3) contributes a nonzero multiple of

$$\delta_{q_0+\ell\log 2} \tag{9.4}$$

and a nonzero multiple of

$$\delta_{q_0-\ell\log 2}. \tag{9.5}$$

No other positive integer has the same positive displacement $\ell \log 2$, and no positive displacement equals its negative. Hence these support points cannot cancel against other terms or against $E\delta_{q_0}$. Equation (9.3) is impossible. □

Remark 9.3 (Relation to exact Heisenberg impossibility)*.* Theorem 2.6 is a universal consequence of the CCR. Proposition 9.2 is more specific: the Möbius Hamiltonian itself does not admit an exactly position-localized stationary mode. The reason is not a measurement uncertainty but the exact translation content of the arithmetic Weyl operators.

10. What is genuinely new in Version 2 and what remains standard

10.1. **Standard ingredients.** The following ingredients are standard and are not presented as new:

(1) the Schrödinger representation of Q and P and the CCR (2.7);
(2) Weyl operators and the Weyl product (3.5);
(3) the arithmetic functions μ, Ω, and classical Möbius inversion;
(4) weighted ℓ^1 Banach-algebra estimates and Neumann inversion.

10.2. **Original notebook constructions.** The following are the constructions/results introduced in this notebook and are the candidates for substantive originality:

(1) the arithmetic phase-space embedding $n \mapsto (\ell \log n, \kappa\Omega(n))$ coupled to the Weyl representation;
(2) the cocycle (3.21) as a deformation of Dirichlet convolution;
(3) the two-sided Heisenberg-twisted Möbius inverse z_ϑ;
(4) the exact prime-ray/prime-pair law (6.1)–(6.3);
(5) the evenness theorem (6.10);
(6) the Möbius Hamiltonian (8.3) and its exact position–momentum equations (8.13)–(8.14);
(7) the pairwise second-order force controlled by $\sin(\Phi_{m,n}/2)$.

Warning 10.1 (Priority). "Introduced in this notebook" is not a bibliographic priority claim. The present version deliberately develops the mathematics first. A separate literature audit must be performed before describing any item above as new in the literature.

11. Open problems

Question 11.1 (Analytic continuation)**.** Does $\zeta_{\mathrm{H}}(s;\vartheta)$ admit a natural meromorphic continuation from $\Re s \geq 2$ to a larger domain for fixed nonzero ϑ?

Question 11.2 (Functional symmetry)**.** Is there a completion of $\zeta_{\mathrm{H}}(s;\vartheta)$ with a symmetry that reduces at $\vartheta = 0$ to the classical reflection $s \mapsto 1 - s$? A nontrivial answer would be required before using the deformation to say anything substantive about the critical line.

Question 11.3 (Spectral zeros)**.** Can the zero set of a continued ζ_{H} be realized as a spectral or resonance set of an operator built intrinsically from $H_{\mu,\sigma,\vartheta}$, rather than as zeros inserted into a multiplier?

Question 11.4 (Classical-zero rigidity)**.** If an analytic two-variable continuation through $\vartheta = 0$ exists, does the exact evenness in ϑ impose any second-order constraint on the motion of the classical Riemann zeros beyond Proposition 7.5?

Question 11.5 (Choice of the second additive coordinate)**.** Which completely additive arithmetic functions $A(n)$, replacing $\Omega(n)$ in $y_n = \kappa A(n)$, produce inequivalent deformations? The choice $A = \Omega$ is distinguished here by its direct relation to the sign of μ on square-free integers, but a classification is open.

12. Notebook history

Remark 12.1. Version 1 used Möbius-weighted logarithmic translations, Fourier multipliers, an eta multiplier in the critical strip, generalized plane-wave null modes, and a standard CCR non-localization statement. Its main defect was that the arithmetic multiplier and the Heisenberg pair were juxtaposed rather than coupled.

Remark 12.2. Version 2 removes the critical-strip multiplier as the central mechanism and replaces ordinary Dirichlet convolution by the Weyl-cocycle product (4.1). Heisenberg noncommutativity now changes the Möbius inverse coefficients themselves and enters the exact equations of motion of the Möbius Hamiltonian.

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